

DSA0601 Data Handling and Visualization

Slot A

CO2 – Assessment Test 1 (AT1)

Chart Construction Test

Instructions:

For each question, write the program in the specified language (R or Python), generate the required

visualization, and answer the interpretation sub-questions clearly with justification.

CO2_AT1 – Chart Construction Test

Integrated Disciplines

Each question combines **Data Visualization + R Programming + another discipline** to promote interdisciplinary learning.

Question 1 (Healthcare Analytics)

Dataset

Patient	Age	BMI	Blood Pressure	Disease Severity
P1	24	19.8	110	Low
P2	32	24.5	118	Medium
P3	41	28.1	132	High
P4	48	30.3	145	High
P5	56	33.2	152	Critical
P6	65	36.5	168	Critical
P7	38	26.2	124	Medium
P8	29	22.1	116	Low

Task

Construct a visualization in **R (ggplot2)** that:

- Maps **Age** to the x-axis.
- Maps **Blood Pressure** to the y-axis.
- Represents **Disease Severity** using an appropriate color palette.
- Represents **BMI** using point size.
- Justify why your aesthetic mappings and scales effectively communicate health risk.

```
library(ggplot2)
```

```
health <- data.frame(
```

```
  Patient = c("P1", "P2", "P3", "P4", "P5", "P6", "P7", "P8"),
```

```
  Age = c(24, 32, 41, 48, 56, 65, 38, 29),
```

```
  BMI = c(19.8, 24.5, 28.1, 30.3, 33.2, 36.5, 26.2, 22.1),
```

```
  BloodPressure = c(110, 118, 132, 145, 152, 168, 124, 116),
```

```
  DiseaseSeverity = factor(
```

```
    c("Low", "Medium", "High", "High", "Critical", "Critical", "Medium", "Low"),
```

```
    levels = c("Low", "Medium", "High", "Critical")
```

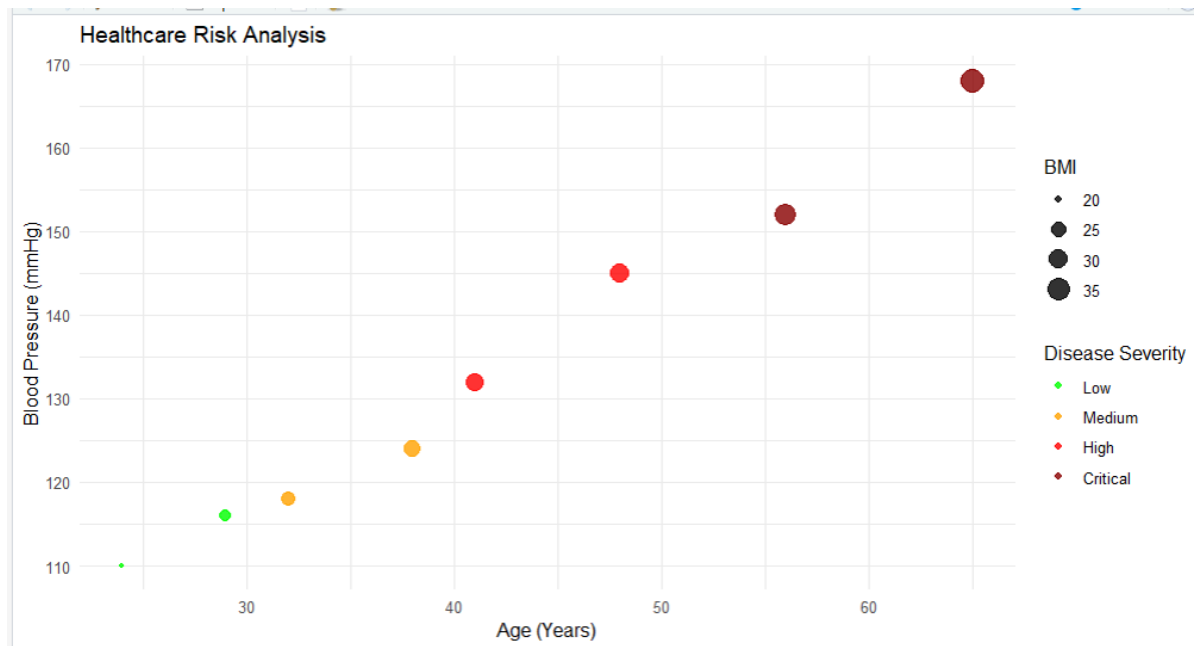
```
)
```

```
)
```

```
ggplot(health, aes(x = Age,
```

```
  y = BloodPressure,
```

```
    color = DiseaseSeverity,  
  
    size = BMI)) +  
  
geom_point(alpha = 0.8) +  
  
scale_color_manual(values = c("green", "orange", "red", "darkred")) +  
  
labs(  
  
  title = "Healthcare Risk Analysis",  
  
  x = "Age (Years)",  
  
  y = "Blood Pressure (mmHg)",  
  
  color = "Disease Severity",  
  
  size = "BMI"  
  
) +  
  
theme_minimal()
```



Justification:

- Age is mapped to the x-axis to show how health risk changes with age.
- Blood Pressure is mapped to the y-axis as it is a key health indicator.
- Disease Severity is represented using colors (Green = Low, Orange = Medium, Red = High, Dark Red = Critical) for easy identification.
- BMI is represented by point size, where larger points indicate higher BMI.
- These mappings help identify high-risk patients by combining multiple health variables in a single visualization.

Interdisciplinary Integration:

- Healthcare: Monitors patient health status.
- Medical Statistics: Analyzes relationships among Age, BMI, Blood Pressure, and Disease Severity.
- Data Visualization: Uses ggplot2 aesthetics to communicate health risk effectively.

Interdisciplinary Integration: Healthcare + Medical Statistics + Data Visualization

Question 2 (Climate Science)

Dataset

Month	Solar Radiation	Temperature (°C)	Rainfall (mm)
Jan	180	20	25
Feb	200	23	18
Mar	240	29	12
Apr	280	35	8
May	320	39	5
Jun	300	36	65
Jul	260	32	150
Aug	250	31	170
Sep	240	30	120
Oct	220	28	85
Nov	190	24	45
Dec	170	21	30

Task

Develop **two visualizations**:

1. A Cartesian line chart showing temperature variation.
2. A Polar coordinate chart representing annual rainfall.

Compare the advantages and limitations of Cartesian and Polar coordinate systems for climate visualization.

```
library(ggplot2)
```

```
climate <- data.frame(
```

```
  Month =
```

```
  factor(c("Jan", "Feb", "Mar", "Apr", "May", "Jun", "Jul", "Aug", "Sep", "Oct", "Nov", "Dec"  
)),
```

```
  levels =
```

```
  c("Jan", "Feb", "Mar", "Apr", "May", "Jun", "Jul", "Aug", "Sep", "Oct", "Nov", "Dec")),
```

```
  SolarRadiation = c(180, 200, 240, 280, 320, 300, 260, 250, 240, 220, 190, 170),
```

```
  Temperature = c(20, 23, 29, 35, 39, 36, 32, 31, 30, 28, 24, 21),
```

```
  Rainfall = c(25, 18, 12, 8, 5, 65, 150, 170, 120, 85, 45, 30)
```

```
)
```

```
ggplot(climate, aes(x = Month, y = Temperature, group = 1)) +
```

```
  geom_line(color = "blue", linewidth = 1.2) +
```

```
  geom_point(color = "red", size = 3) +
```

```
  labs(
```

```
    title = "Monthly Temperature Variation",
```

```
    x = "Month",
```

```
    y = "Temperature (°C)"
```

```
  ) +
```

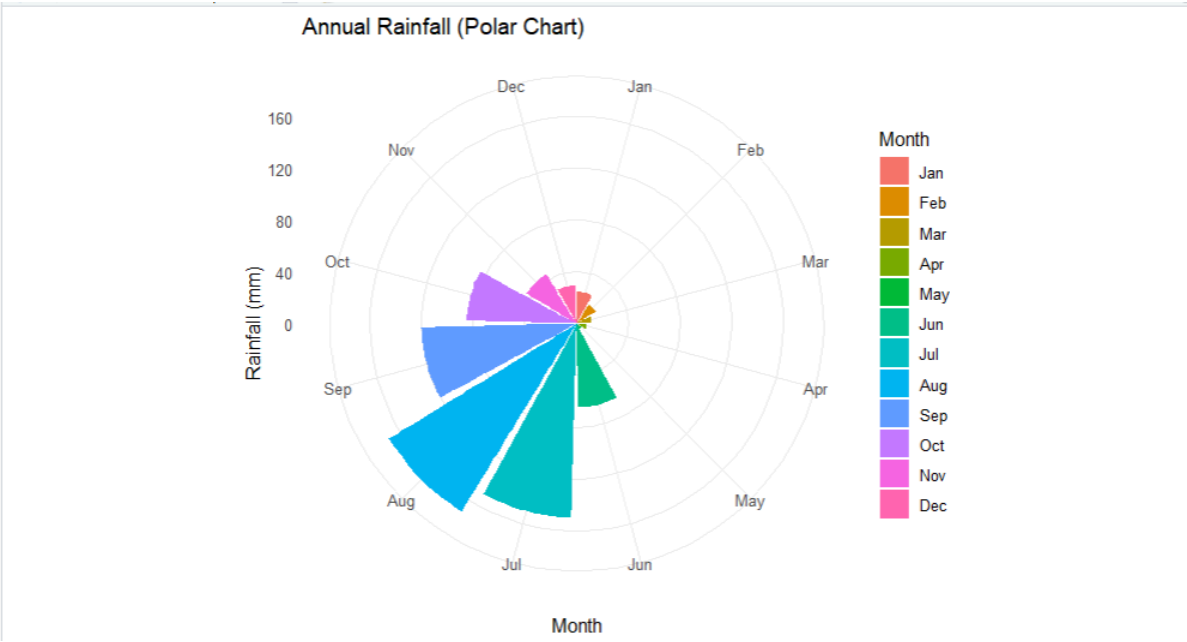
```
  theme_minimal()
```

```
ggplot(climate, aes(x = Month, y = Rainfall, fill = Month)) +
```

```
  geom_bar(stat = "identity") +
```

```
  coord_polar() +
```

```
labs(  
  title = "Annual Rainfall (Polar Chart)",  
  x = "Month",  
  y = "Rainfall (mm)"  
) +  
theme_minimal()
```



Comparison of Cartesian and Polar Coordinate Systems

Cartesian Coordinate System	Polar Coordinate System
Best for showing trends over time.	Best for showing cyclic or seasonal patterns.
Easy to compare exact values and changes.	Highlights yearly distribution in a circular format.
Simple to interpret and analyze.	Attractive visualization but harder to compare precise values.

Suitable for temperature trends.	Suitable for annual rainfall patterns.
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Interdisciplinary Integration

- **Environmental Science:** Studies monthly temperature and rainfall patterns.
- **Mathematics:** Uses Cartesian and Polar coordinate systems to represent data.
- **Data Visualization:** Employs line and polar charts to communicate climate information effectively.

Interdisciplinary Integration: Environmental Science + Mathematics + Data Visualization
Question 3 (Business Analytics & Marketing)

Dataset

Product	Sales (₹ Lakhs)	Profit (%)	Market Share (%)
A	210	18	24
B	185	14	20
C	260	25	28
D	145	12	15
E	120	10	13

Task

Construct a visualization where:

- Sales are represented by bar height.
 - Profit percentage is represented using a sequential color scale.
 - Highlight the product with the highest market share using a contrasting color.
- Explain why your chosen color palette improves business decision-making.

```
library(ggplot2)
```

```

business <- data.frame(

  Product = c("A","B","C","D","E"),

  Sales = c(210,185,260,145,120),

  Profit = c(18,14,25,12,10),

  MarketShare = c(24,20,28,15,13)

)


business$Highlight <- ifelse(

  business$MarketShare == max(business$MarketShare),

  "Highest Market Share",

  as.character(business$Profit)

)


ggplot(business, aes(x = Product, y = Sales, fill = Highlight)) +

  geom_bar(stat = "identity") +

  scale_fill_manual(

    values = c(

      "10" = "#deebf7",

      "12" = "#c6dbef",

      "14" = "#9ecae1",

```

```

"18" = "#6baed6",

"Highest Market Share" = "red"

)

)+

labs(

  title = "Sales, Profit and Market Share Analysis",

  x = "Product",

  y = "Sales (₹ Lakhs)",

  fill = "Profit / Highlight"

)+

theme_minimal()

```



Why this Color Palette Improves Business Decision-Making

- A sequential blue color scale represents increasing profit percentages, making profit differences easy to identify.
- The red color highlights the product with the highest market share, drawing immediate attention.
- Decision-makers can quickly compare sales, profit, and market share in one visualization.
- The contrasting colors help identify top-performing products and support better business strategies.

Interdisciplinary Integration

- **Business Analytics:** Evaluates sales, profit, and market share.
- **Marketing:** Identifies the leading product for strategic planning.
- **Data Visualization:** Uses bar height and color to communicate business performance effectively.

Interdisciplinary Integration: Marketing + Business Intelligence + Visualization

Question 4 (Transportation Engineering)

Dataset

Direction	Vehicles/hour
North	980
North-East	760
East	1240
South-East	890
South	1420
South-West	1130
West	860

Direction	Vehicles/hour
North-West	720

Task

Using **R Programming**:

- Construct a **Polar Coordinate Chart** representing traffic flow.
- Apply an appropriate color palette to distinguish traffic density.
- Explain why a Polar chart is more suitable than a Cartesian bar chart for directional traffic analysis.

```
library(ggplot2)
```

```
traffic <- data.frame(
```

```

Direction = factor(

  c("North", "North-East", "East", "South-East", "South", "South-West", "West", "North-
    West"),

  levels =
    c("North", "North-East", "East", "South-East", "South", "South-West", "West", "North-
      West")

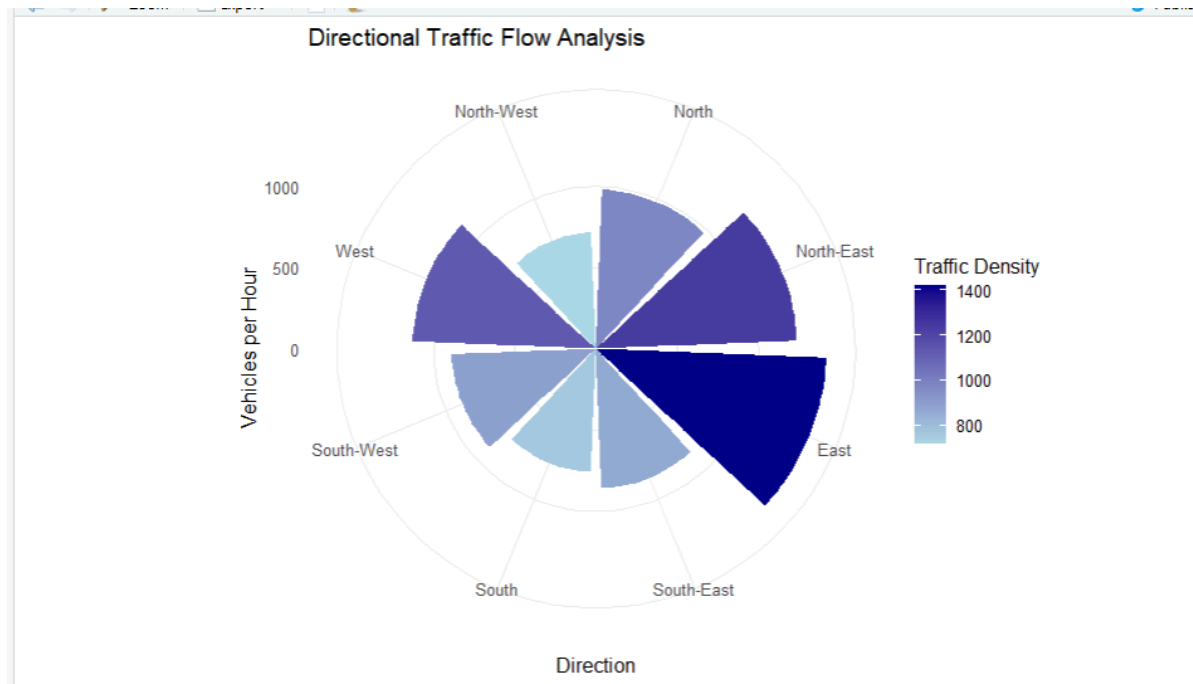
),

Vehicles = c(980,1240,1420,860,760,890,1130,720)

)

ggplot(traffic, aes(x = Direction, y = Vehicles, fill = Vehicles)) +
  geom_bar(stat = "identity") +
  coord_polar() +
  scale_fill_gradient(low = "lightblue", high = "darkblue") +
  labs(
    title = "Directional Traffic Flow Analysis",
    x = "Direction",
    y = "Vehicles per Hour",
    fill = "Traffic Density"
  ) +
  theme_minimal()

```



Why a Polar Chart is More Suitable

- A polar chart naturally represents directional data such as North, East, South, and West.
- It makes it easier to compare traffic flow from different directions around a central point.
- The circular layout reflects real-world road intersections and traffic movement.
- A Cartesian bar chart shows values clearly but does not effectively represent direction.

Interdisciplinary Integration

- **Civil Engineering:** Analyzes traffic flow for road planning.
- **GIS:** Represents directional traffic patterns spatially.
- **Transportation Analytics:** Visualizes traffic density to support traffic management and infrastructure decisions.

Interdisciplinary Integration: Civil Engineering + GIS + Transportation Analytics

Question 5 (Environmental Sustainability Dashboard)

Dataset

City	AQI	Green Cover (%)	CO ₂ Emission	Population (Lakhs)
A	48	45	1.8	9
B	82	32	3.5	15
C	135	21	5.6	22
D	165	18	7.4	28
E	72	38	2.9	12
F	95	30	4.2	18

Task

Design a visualization dashboard that includes:

- Appropriate aesthetic mappings.
- Suitable scales for continuous variables.
- An effective color palette for AQI categories.
- Justification for every visualization design choice.

```
library(ggplot2)
```

```
library(gridExtra)
```

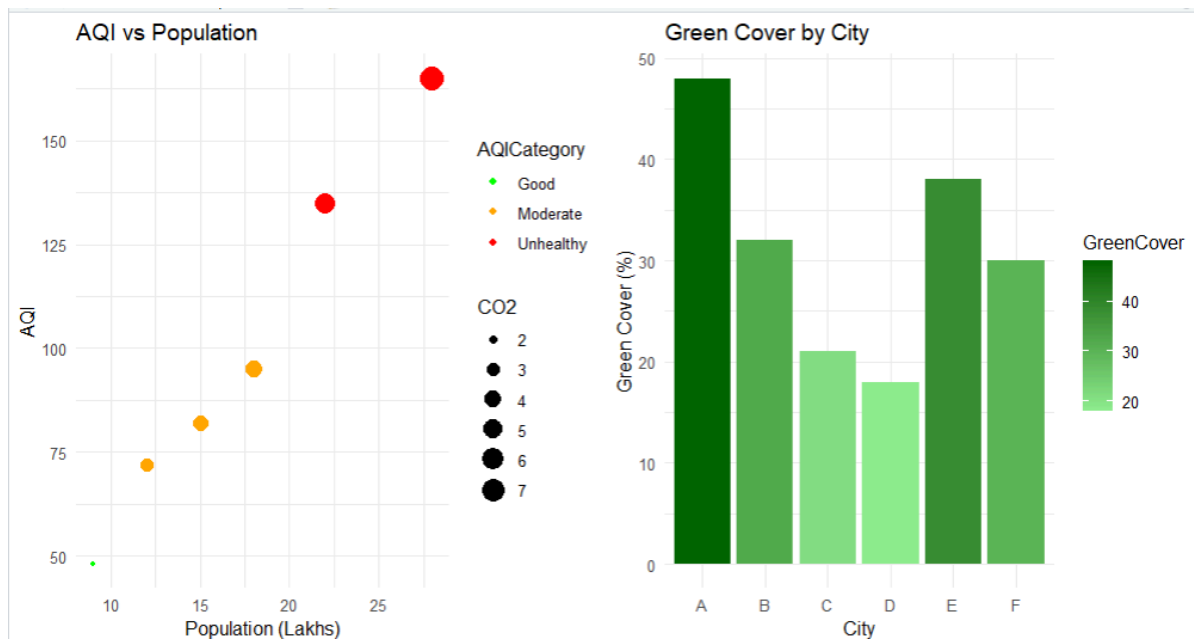
```
environment <- data.frame(
  City = c("A","B","C","D","E","F"),
  AQI = c(48,82,135,165,72,95),
  GreenCover = c(48,32,21,18,38,30),
  CO2 = c(1.8,3.5,5.6,7.4,2.9,4.2),
  Population = c(9,15,22,28,12,18)
)
```

```
environment$AQICategory <- factor(
  c("Good", "Moderate", "Unhealthy", "Unhealthy", "Moderate", "Moderate"),
  levels = c("Good", "Moderate", "Unhealthy")
)
```

```
p1 <- ggplot(environment, aes(x = Population, y = AQI,
                              color = AQICategory,
                              size = CO2)) +
  geom_point() +
  scale_color_manual(values = c("green", "orange", "red")) +
  labs(title = "AQI vs Population",
        x = "Population (Lakhs)",
        y = "AQI") +
  theme_minimal()
```

```
p2 <- ggplot(environment, aes(x = City, y = GreenCover, fill = GreenCover)) +
  geom_bar(stat = "identity") +
  scale_fill_gradient(low = "lightgreen", high = "darkgreen") +
  labs(title = "Green Cover by City",
        x = "City",
        y = "Green Cover (%)") +
  theme_minimal()
```

```
grid.arrange(p1, p2, ncol = 2)
```



Justification for Visualization Design Choices

- **Scatter Plot:** Population is mapped to the x-axis, AQI to the y-axis, CO₂ emission to point size, and AQI category to color, allowing multiple environmental factors to be analyzed together.
- **Bar Chart:** Green Cover is represented using bar height for easy comparison across cities.
- **Continuous Scales:** Gradient scales effectively represent variations in Green Cover and CO₂ emissions.
- **Color Palette:** Green indicates good air quality, orange indicates moderate air quality, and red indicates unhealthy air quality, making AQI categories easy to interpret.
- **Dashboard Layout:** Combining two visualizations provides a comprehensive view of environmental sustainability indicators.

Interdisciplinary Integration

- **Environmental Sustainability:** Monitors air quality, green cover, and CO₂ emissions.
- **Urban Planning:** Evaluates environmental conditions across cities.
- **Data Visualization:** Uses multiple charts, aesthetic mappings, and color scales to present sustainability data effectively.